

CS2422 Assembly Language & System Programming

September 30, 2003

Today's Topics

- Arithmetic Instructions
- Jump and Loop

Study Guide

- Section 4.2: Arithmetic Instructions
- Section 4.5: JMP and LOOP

Addition and Subtraction

- ADD X, Y
 $X := X + Y$
- SUB X, Y
 $X := X - Y$

INC, DEC, NEG

- INC X
X := X + 1 or X++
- DEC X
X := X - 1 or X--
- NEG X
X := -X

Expression

- Example: $X = (A + B) * (D - E)$

MOV	EAX, A
ADD	EAX, B
MOV	ECX, D
SUB	ECX, E
IMUL	EAX, ECX
MOV	X, EAX

Flags Affected

- Flags (register) tell us whether any of the following conditions occur:
 - Overflow,
 - Carry,
 - Zero, Sign...etc.
- Used for decision in branch.
 - Loop (discussed next)
 - If...then...else

Zero and Sign

- Zero Flag ZF=1 if the instruction produce 0.
MOV CX, 1
SUB CX, 1 ; CX=0, ZF=1
- Sign Flag SF=1 if the instruction produce a negative number.
MOV CX, 0
SUB CX, 1 ; CX=-1, SF=1
ADD CX, 2 ; CX=1, SF=0

Carry (Unsigned Arithmetic)

- Example:

```
MOV AL, 0FFh
```

```
ADD AL, 1 ; CF = 1, AL=00
```

```
MOV AX, 00FFh
```

```
ADD AX, 1 ; CF = 0, AX=0100h
```

Overflow (Signed Arithmetic)

- Example:

```
MOV AL, +127
```

```
ADD AL, 1 ; OF = 1
```

```
MOV AL, -128
```

```
SUB AL, 1 ; OF = 1
```

Detecting Carry

- Detecting Carry is easy.
 - Adding two N-bit numbers result in an (N+1)-bit number.
- Example:

```
  0 0 0 0 0 1 0 0
+  1 1 1 1 1 1 1 1
1  0 0 0 0 0 0 1 1
```
- CF is ignored for signed arithmetic. For example, the above is $4 + (-1)$ in decimal

Detecting Overflow

- Carry isn't meaningful for signed arithmetic. For example, adding any two negative numbers always produces carry.
- Detecting Overflow:
 - Compare CF and MSB (Most Significant Bit).
 - See textbook page 112 for more detail.

Overflow in Positive Numbers

- Carry never happens. For example:

0 1 1 1 1 1 1 1 (127)

+ 0 1 1 1 1 1 1 1 (127)

- Overflow occurs if MSB becomes 1
 - MSB=1 indicates a negative number.
 - But, we're adding two positive numbers...?!

Overflow in Negative Numbers

- Carry always happens. For example:

1 0 0 0 0 0 0 0 (-128)

+ 1 1 1 1 1 1 1 1 (-1)

- Overflow occurs if MSB becomes 0
 - MSB=0 indicates a positive number.
 - But, we're adding two negative numbers...?!

Implementation of Loops

- JMP instruction: Unconditional Branch.
- LOOP instruction:
 - Step 1: Set ECX to n for a loop of n iterations.
 - Step 2: Use LOOP instruction at the end of loop.
 - Hidden action: DEC ECX

Example 1: Summation

- For I := 10 downto 1 {Sum := Sum+I}

```
MOV ECX, 10
MOV EAX, 0
L1:  ADD EAX, ECX
     LOOP L1
```

Example 2: Summation

- For I := 1 to 10 {Sum := Sum+I}

```
MOV ECX, 10
MOV EAX, 0
MOV EDX, 1
L1: ADD EAX, EDX
    INC EDX
    LOOP L1
```

Example 3: Array Traversal

- Exercise: what is computed and stored at EAX?

```
MOV ECX, 10
MOV EAX, 0
MOV EDI, OFFSET var1
L1: ADD EAX, [EDI]
    INC EDI
    LOOP L1
```